

Regularity in the Calculus of Variations

Assignment I

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1 Problem 1

Let $\Omega = B_1^n(0)$ and $u(x) = |x|^{-s}$ with $s > 0$.

1.1 Part 1, Case 1: $u \in L^q(\Omega)$

This must satisfy the finite q -integrability of the function $u(x)$, that is:

$$\int_{\Omega} |u(x)|^q dx < \infty.$$

We can determine the condition for the exponent s from the above integration criteria in L^q space. The only concern for this integration is that it will be extremely cumbersome to integrate in cartesian coordinate system. Now, the domain is given as n -dimensional unit ball ($\Omega = B_1^n(0)$). Thus, the accurate coordinate system is spherical polar coordinates. It will decompose the measure in radial measure ($r = |x|$) and angular measure ($\omega \in S^{n-1}$). The usual integration in $B_1^n(0)$ for any function $f(x)$ is in the following form:

$$\int_{B_1^n(0)} f(x) dx = \int_{r=0}^1 \int_{\omega \in S^{n-1}} f(r\omega) r^{n-1} d\sigma(\omega) dr.$$

Applying this integration for $u(x)$ gives:

$$\int_{\Omega} |u(x)|^q dx = \int_0^1 \int_{S^{n-1}} |u(r\omega)|^q r^{n-1} d\sigma dr. \quad (1)$$

Putting $u(x) = |x|^{-s}$ and $u(r\omega) = r^{-s}$ in (1), gives:

$$\int_{\Omega} |x|^{-sq} dx = \int_0^1 \int_{S^{n-1}} r^{-sq} r^{n-1} d\sigma dr.$$

Decomposing the above equation gives:

$$\int_{\Omega} |x|^{-sq} dx = \left(\int_{S^{n-1}} d\sigma \right) \int_0^1 r^{n-1-sq} dr. \quad (2)$$

Now, the first integration gives surface measure on n -sphere:

$$|S^{n-1}| = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})},$$

where $\Gamma(\cdot)$ is the usual gamma function. So (2) becomes:

$$\int_{\Omega} |x|^{-sq} dx = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} \int_0^1 r^{n-1-sq} dr. \quad (3)$$

The main objective here is that the integration expression of (3) must be finite for $u(x) \in L^q$.

Now $\frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})} < \infty$ as the domain is finite dimensional. Hence we need:

$$\int_0^1 r^{n-1-sq} dr < \infty.$$

This will converge when:

$$n - 1 - sq > -1.$$

So the condition of s for $u(x) \in L^q(\Omega)$ is:

$$\boxed{n > sq.}$$

1.2 Case 2: $u \in W^{1,p}(\Omega)$

From the definition of Sobolev space:

$$u \in W^{1,p} \iff u \in L^p(\Omega), \nabla u \in L^p(\Omega).$$

Case 2(a): $u \in L^p(\Omega)$

It will give the similar condition as $u \in L^q(\Omega)$, with q replaced by p :

$$\boxed{n > sp.}$$

Case 2(b): $\nabla u \in L^p(\Omega)$

The weak derivative expression of $u(x)$ is:

$$\nabla u = -s|x|^{-s-2}x.$$

So, the absolute value of the derivative is:

$$|\nabla u(x)| = s|x|^{-s-1}. \quad (4)$$

$\nabla u \in L^p(\Omega) \implies :$

$$\int_{\Omega} |\nabla u|^p dx = \frac{2\pi^{n/2}s}{\Gamma(\frac{n}{2})} \int_0^1 r^{-p(s+1)} r^{n-1} dr < \infty.$$

That is:

$$\int_{\Omega} |\nabla u|^p dx = \frac{2\pi^{n/2}s}{\Gamma(\frac{n}{2})} \int_0^1 r^{n-1-p(s+1)} dr.$$

This p -integrability will be finite only if:

$$\int_0^1 r^{n-1-p(s+1)} dr < \infty.$$

This integration will converge if:

$$n - 1 - p(s + 1) > -1.$$

Which gives:

$$s < \frac{n}{p} - 1.$$

Hence, the condition of s for $u \in W^{1,p}(\Omega)$ is:

$$\boxed{n > sp \quad \text{and} \quad s < \frac{n}{p} - 1.}$$

Since $u \in C^1(\Omega \setminus \{0\})$, the singular set $\{0\}$ has measure zero, and both u and its classical gradient belong to $L^p(\Omega)$, it follows that the classical gradient coincides with the weak gradient. Hence, ∇u is the weak derivative and $u \in W^{1,p}(\Omega)$.

1.3 Part 2: Sharp Sobolev exponent p^*

For $1 \leq p < n$, Sobolev embedding theorem gives that:

$$W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega),$$

where $p^* = \frac{np}{n-p}$ and satisfies the Sobolev inequality:

$$\|u\|_{L^{p^*}} \leq C \|u\|_{W^{1,p}}.$$

Sharp Sobolev exponent means p^* cannot be replaced by any larger exponent than p^* . This implies:

$$W^{1,p}(\Omega) \not\hookrightarrow L^q(\Omega)$$

for any $q > p^*$. From the previous finding condition for $u \in W^{1,p}(\Omega)$:

$$s < \frac{n}{p} - 1.$$

Taking $\epsilon > 0$ gives:

$$s = \frac{n}{p} - 1 - \epsilon. \tag{5}$$

Putting (5) in the inequality $sq < n$ (from the condition of s for $u \in L^q(\Omega)$):

$$\left(\frac{n}{p} - 1 - \epsilon\right) q < n.$$

Taking $\epsilon \rightarrow 0$ gives:

$$q \leq \frac{np}{n-p} = p^*.$$

This proves that the Sobolev exponent p^* is sharp.

2 Problem 2

It is given that the dimension $n \geq 2$. The function $u(x) = \frac{x}{|x|} \in B_1^n(0)$. We need to find the range of p such that $u \in W^{1,p}(B_1^n(0))$. From the definition of Sobolev space:

$$u \in W^{1,p}(B_1^n(0)) \iff u \in L^p(B_1^n(0)), \nabla u \in L^p(B_1^n(0)).$$

2.1 Case 1: $u \in L^p(B_1^n(0))$

$u(x) = \frac{x}{|x|}$ for all $x \in B_1^n(0) \setminus \{0\}$:

$$|u(x)| = \left| \frac{x}{|x|} \right| = \frac{|x|}{|x|} = 1 \quad \text{for all } x \neq 0.$$

Hence, for any $p \geq 1$:

$$|u(x)|^p = 1.$$

Thus, the p -th integration in spherical polar coordinates is:

$$\int_{B_1^n(0)} |u(x)|^p dx = \int_{B_1^n(0)} dx = |B_1^n(0)|.$$

Now, as the unit ball is in finite dimension, it will always be bounded. This says:

$$\int_{B_1^n(0)} |u(x)|^p dx < \infty \quad \text{for every } 1 \leq p \leq \infty.$$

2.2 Case 2: $\nabla u \in L^p(B_1^n(0))$

For $u(x) = \frac{x}{|x|}$, the absolute value of the weak derivative is:

$$|\nabla u(x)| = \frac{1}{|x|}.$$

Now, the L^p integrability in spherical polar coordinates gives:

$$\int_{B_1^n(0)} |\nabla u(x)|^p dx = S_M \int_0^1 r^{n-1-p} dr,$$

where $S_M = \frac{2\pi^{n/2}}{\Gamma(\frac{n}{2})}$ is the surface measure, which is finite. This integrability will converge $\iff$:

$$n - 1 - p > -1.$$

$$\iff p < n.$$

Hence the range of p for $u \in W^{1,p}(B_1^n(0))$ is:

$$\boxed{1 \leq p < n.}$$

Since $u \in C^1(\Omega \setminus \{0\})$, the singular set $\{0\}$ has measure zero, and both u and its classical gradient belong to $L^p(\Omega)$, it follows that the classical gradient coincides with the weak gradient. Hence, ∇u is the weak derivative and $u \in W^{1,p}(B_1^n(0))$.

2.3 Part 2: $n = 1$

Proof. For $n = 1$, take the domain as an open interval (a, b) . Let $u \in W^{1,p}(a, b)$. Then, for almost every $x, y \in (a, b)$, by the fundamental theorem of calculus:

$$u(x) - u(y) = \int_x^y u'(t) dt. \quad (1)$$

Taking absolute value on (1) and using Jensen's inequality:

$$|u(x) - u(y)| = \left| \int_x^y u'(t) dt \right| \leq \int_y^x |u'(t)| dt.$$

Using Hölder:

$$|u(x) - u(y)| \leq |x - y|^{1-1/p} \|u'\|_{L^p(a,b)}.$$

Hence:

$$|u(x) - u(y)| \leq C|x - y|^\alpha \quad \text{with } \alpha = 1 - \frac{1}{p} > 0.$$

This is the Hölder estimate. Hence:

$$u \in C^{0,\alpha}([a, b]) \subset C([a, b]).$$

Thus, for $n = 1$, every Sobolev function is continuous. The above proof showed u is Hölder continuous (for $p > 1$) or absolutely continuous (for $p = 1$). $\square$

3 Problem 3

Proof. The given function is:

$$u(x) = |\log |x||^s.$$

The domain is $B_{1/2}^2(0)$. Hence the dimension is $n = 2$. From the definition of Sobolev function:

$$u(x) \in W^{1,2}(B_{1/2}^2(0)) \iff u \in L^2(B_{1/2}^2(0)), \nabla u(x) \in L^2(B_{1/2}^2(0)).$$

So, this problem is strictly about $p = 2$ for 2D.

3.1 Case 1: $u(x) \in L^2(B_{1/2}^2(0))$

This is a radial function:

$$u(x) = |\log r|^s.$$

The condition for square integrability is:

$$\int_{B_{1/2}^2(0)} |u|^2 dx < \infty.$$

The integration of the radial function in 2D spherical polar coordinates is:

$$\int_{B_{1/2}^2(0)} |u|^2 dx = \int_0^{1/2} \int_{S^1} |\log r|^{2s} r d\sigma dr. \quad (6)$$

The surface measure gives $|S^1| = 2\pi$. Hence, (6) becomes:

$$\int_{B_{1/2}^2(0)} |u|^2 dx = 2\pi \int_0^{1/2} |\log r|^{2s} r dr. \quad (7)$$

Now, make the change of variables:

$$\begin{aligned} t &= -\log r, \\ r &= e^{-t}, \\ dr &= -e^{-t} dt. \end{aligned}$$

The lower integration limit becomes $\log 2$ and upper integration limit is ∞ . Hence (7) becomes:

$$\int_{B_{1/2}^2(0)} |u|^2 dx = 2\pi \int_{\log 2}^{\infty} t^{2s} e^{-2t} dt.$$

Make another change of variables: $z = 2t$, so $t = z/2$, $dt = dz/2$:

$$\int_{B_{1/2}^2(0)} |u|^2 dx = \frac{2\pi}{2^{2s+1}} \int_{2\log 2}^{\infty} z^{2s} e^{-z} dz.$$

That is:

$$\int_{B_{1/2}^2(0)} |u|^2 dx = \frac{\pi}{2^{2s}} \Gamma(2s+1, 2\log 2), \quad (8)$$

where $\Gamma(\cdot, \cdot)$ is the upper incomplete gamma function. The prefactor is finite, and $\Gamma(2s+1, 2\log 2)$ is finite for all $s > -1/2$. Since $0 < s < 1$, the integral is finite:

$$\int_{B_{1/2}^2(0)} |u|^2 dx < \infty.$$

This proves:

$$\boxed{u(x) \in L^2(B_{1/2}^2(0)).}$$

3.2 Case 2: $\nabla u \in L^2(B_{1/2}^2(0))$

The derivative of the function is:

$$|\nabla u(x)| = s |\log r|^{s-1} \frac{1}{r}.$$

Now, test square-integrability:

$$\int_{B_{1/2}^2(0)} |\nabla u(x)|^2 dx = \int_0^{1/2} \int_{S^1} s^2 |\log r|^{2s-2} \frac{1}{r^2} r d\sigma dr.$$

With further simplification:

$$\int_{B_{1/2}^2(0)} |\nabla u(x)|^2 dx = 2\pi s^2 \int_0^{1/2} |\log r|^{2s-2} \frac{1}{r} dr. \quad (9)$$

Make the change of variables:

$$\begin{aligned} t &= -\log r, \\ \frac{1}{r} dr &= -dt. \end{aligned}$$

This gives lower limit $\log 2$ and upper limit ∞ :

$$\int_{B_{1/2}^2(0)} |\nabla u(x)|^2 dx = 2\pi s^2 \int_{\log 2}^{\infty} t^{2s-2} dt.$$

This integral converges if and only if:

$$2s - 2 < -1 \iff s < \frac{1}{2}.$$

The case $s = 0$ gives $u(x) = 1$, which is constant and hence $u \in W^{1,2}$ (trivial case).

Thus:

$$\boxed{u \in W^{1,2}(B_{1/2}^2(0)) \quad \text{for all } 0 \leq s < 1/2.}$$

Since $u \in C^1(B_{1/2}^2(0) \setminus \{0\})$ and both u and ∇u belong to $L^2(B_{1/2}^2(0))$, the classical gradient coincides with the weak gradient. The singular set $\{0\}$ has measure zero. $\square$

4 Problem 4

Proof. Let $n \geq 2$ and let $U = B_1^n(0) \subset \mathbb{R}^n$. Define

$$u(x) = \log(\log(1 + \frac{1}{|x|})).$$

We shall show that $u \in W^{1,n}(U)$, i.e.,

$$u \in L^n(U) \quad \text{and} \quad \nabla u \in L^n(U).$$

Case 1: $u \in L^n(U)$

Since u is smooth away from $x = 0$, the only possible singularity is at the origin. As $r = |x| \rightarrow 0$,

$$u(x) = \log(\log(1 + 1/r)) \sim \log(\log(1/r)),$$

which grows slower than any power of r^{-1} . Hence

$$\int_{B_1(0)} |u(x)|^n dx < \infty,$$

and therefore $u \in L^n(U)$.

Case 2: $\nabla u \in L^n(U)$

Let $r = |x|$ and define

$$f(r) = \log(\log(1 + 1/r)).$$

Then $u(x) = f(|x|)$ and by the chain rule,

$$f'(r) = \frac{1}{\log(1 + 1/r)} \cdot \frac{1}{1 + 1/r} \cdot \left(-\frac{1}{r^2}\right) = -\frac{1}{r(r+1)\log(1 + 1/r)}.$$

Since

$$\nabla u(x) = f'(r) \frac{x}{r},$$

we obtain

$$|\nabla u(x)| = |f'(r)| = \frac{1}{r(r+1)\log(1 + 1/r)}.$$

We compute:

$$\begin{aligned} \int_{B_1^n(0)} |\nabla u(x)|^n dx &= C_n \int_0^1 \frac{1}{r^n(r+1)^n(\log(1 + 1/r))^n} r^{n-1} dr \\ &= C_n \int_0^1 \frac{1}{r(\log(1 + 1/r))^n} \cdot \frac{1}{(r+1)^n} dr. \end{aligned}$$

Since $0 < r < 1$, we have $(r+1)^{-n} \leq 1$, hence

$$\int_{B_1^n(0)} |\nabla u|^n dx \leq C_n \int_0^1 \frac{dr}{r(\log(1 + 1/r))^n}.$$

Set

$$t = \log(1 + 1/r).$$

Then

$$dt = -\frac{1}{r(r+1)} dr, \quad \text{hence} \quad \frac{dr}{r} = -(r+1) dt \leq 2 dt.$$

Changing variables and reversing the limits of integration:

$$\begin{aligned} \int_0^1 \frac{dr}{r(\log(1 + 1/r))^n} &= \int_{\infty}^{\log 2} \frac{-(r+1)}{t^n} dt \\ &= \int_{\log 2}^{\infty} \frac{r+1}{t^n} dt. \end{aligned}$$

Since $0 < r < 1$ implies $1 < r+1 < 2$, we have:

$$\int_0^1 \frac{dr}{r(\log(1 + 1/r))^n} \leq 2 \int_{\log 2}^{\infty} \frac{dt}{t^n}.$$

The integral on the right converges if and only if $n > 1$, which holds by assumption (indeed $n \geq 2$).

Thus,

$$\int_{B_1^n(0)} |\nabla u|^n dx < \infty.$$

We have shown that

$$u \in L^n(B_1^n(0)) \quad \text{and} \quad \nabla u \in L^n(B_1^n(0)).$$

Hence

$$u \in W^{1,n}(B_1^n(0)).$$

□

5 Problem 5

We have to prove the following theorem:

Theorem 5.1. *For every $1 \leq p < \infty$, there exists a constant $c = c(n, p) > 0$ such that*

$$\oint_{B_R(x)} |u - (u)_{B_R(x)}|^p \leq cR^p \oint_{B_R(x)} |\nabla u|^p$$

for any ball $B_R(x) \subset \mathbb{R}^n$ and every $u \in W^{1,p}(\mathbb{R}^n)$.

Proof. Here, $(u)_{B_R(x)}$ represents the average or mean integral of u over $B_R(x)$:

$$(u)_{B_R(x)} := \frac{1}{|B_R(x)|} \int_{B_R(x)} u(y) dy = \oint_{B_R(x)} u(y) dy.$$

Initially, assume $u \in C^1(\mathbb{R}^n)$. Then:

$$\oint_{B_R(x)} |u(y) - (u)_{B_R(x)}|^p dy = \oint_{B_R(x)} \left| \oint_{B_R(x)} (u(y) - u(w)) dw \right|^p dy. \quad (10)$$

This is valid because:

$$\oint_{B_R(x)} (u(y) - u(w)) dw = u(y) - \oint_{B_R(x)} u(w) dw = u(y) - (u)_{B_R(x)}.$$

Applying Jensen's inequality to the right side of (9):

$$\oint_{B_R(x)} \left| \oint_{B_R(x)} (u(y) - u(w)) dw \right|^p dy \leq \oint_{B_R(x)} \oint_{B_R(x)} |u(y) - u(w)|^p dy dw.$$

Thus:

$$\oint_{B_R(x)} |u(y) - (u)_{B_R(x)}|^p dy \leq \oint \oint_{B_R(x)} |u(y) - u(w)|^p dy dw. \quad (11)$$

We use the following lemma:

Lemma 5.2. *For every $1 \leq p < \infty$, there exists $c > 0$ depending only on n and p such that*

$$\int_{B_R(x)} |u(y) - u(w)|^p dy \leq cR^{p-1} \int_{B_R(x)} \frac{|\nabla u(y)|^p}{|y - w|^{n-1}} dy$$

for every ball $B_R(x) \subset \mathbb{R}^n$, every $w \in B_R(x)$, and every $u \in W^{1,p}(\mathbb{R}^n)$.

Using this lemma to estimate the right side of (10):

$$\oint \oint_{B_R(x)} |u(y) - u(w)|^p dy dw \leq c \oint_{B_R(x)} R^{p-1} \int_{B_R(x)} \frac{|\nabla u(w)|^p}{|y - w|^{n-1}} dw dy. \quad (12)$$

There is no n in the exponent of R in (11) due to the mean integral. Since $|y - w| \leq 2R$:

$$\int_{B_R(x)} \frac{|\nabla u(w)|^p}{|y - w|^{n-1}} dw dy \leq CR \int_{B_R(x)} |\nabla u(w)|^p dw.$$

Hence:

$$\oint_{B_R(x)} |u(y) - (u)_{B_R(x)}|^p dy \leq cR^{p-1} \oint_{B_R(x)} R \int_{B_R(x)} |\nabla u(w)|^p dw \quad (13)$$

$$= cR^p \oint_{B_R(x)} |\nabla u(w)|^p dw. \quad (14)$$

Thus:

$$\boxed{\oint_{B_R(x)} |u - (u)_{B_R(x)}|^p \leq cR^p \oint_{B_R(x)} |\nabla u|^p.}$$

The result extends to $u \in W^{1,p}(\mathbb{R}^n)$ by density. $\square$

6 Problem 6

Theorem 6.1 (Poincaré inequality on convex domains). *Let $\Omega \subset \mathbb{R}^n$ be open, bounded, convex and C^1 . Then for every $1 \leq p < \infty$ there exists a constant $c = c(n, p, \Omega) > 0$ such that*

$$\oint_{\Omega} |u - (u)_{\Omega}|^p dx \leq c \oint_{\Omega} |\nabla u|^p dx \quad \text{for all } u \in W^{1,p}(\Omega).$$

Proof. Fix $u \in W^{1,p}(\Omega)$. Since Ω is convex, for every $x, y \in \Omega$ the line segment $[x, y] \subset \Omega$, and by the fundamental theorem of calculus,

$$u(x) - u(y) = \int_0^1 \nabla u(x + t(y - x)) \cdot (y - x) dt.$$

Taking absolute values:

$$|u(x) - u(y)| \leq |x - y| \int_0^1 |\nabla u(x + t(y - x))| dt. \quad (13)$$

Using the identity

$$u(x) - (u)_{\Omega} = \oint_{\Omega} (u(x) - u(y)) dy,$$

and applying (13), we obtain:

$$|u(x) - (u)_{\Omega}| \leq \oint_{\Omega} |x - y| \int_0^1 |\nabla u(x + t(y - x))| dt dy.$$

Interchanging the integrals:

$$|u(x) - (u)_\Omega| \leq \int_0^1 \int_\Omega |x - y| |\nabla u(x + t(y - x))| dy dt. \quad (14)$$

Since $|x - y| \leq \text{diam}(\Omega)$, (14) becomes:

$$|u(x) - (u)_\Omega| \leq \text{diam}(\Omega) \int_0^1 \int_\Omega |\nabla u(x + t(y - x))| dy dt. \quad (15)$$

Raising to the power p , integrating in x , and applying Jensen's inequality:

$$\int_\Omega |u - (u)_\Omega|^p dx \leq C(\Omega) \int_\Omega \int_0^1 \int_\Omega |\nabla u(x + t(y - x))|^p dy dt dx, \quad (16)$$

where $C(\Omega) = \text{diam}(\Omega)^p$.

For fixed $t \in (0, 1)$ and $y \in \Omega$, the map

$$x \mapsto z = x + t(y - x)$$

maps Ω into itself and is Lipschitz with Jacobian bounded independently of x, y, t . Hence:

$$\int_\Omega |\nabla u(x + t(y - x))|^p dx \leq C(n) \int_\Omega |\nabla u(z)|^p dz.$$

Substituting into (16):

$$\int_\Omega |u - (u)_\Omega|^p dx \leq C(\Omega) C(n) \int_0^1 \int_\Omega |\nabla u(z)|^p dz dt.$$

Since $\int_0^1 dt = 1$, we conclude:

$$\int_\Omega |u - (u)_\Omega|^p dx \leq C(\Omega) C(n) \int_\Omega |\nabla u|^p dx.$$

Dividing by $|\Omega|$:

$$\int_\Omega |u - (u)_\Omega|^p dx \leq c(n, p, \Omega) \int_\Omega |\nabla u|^p dx,$$

where

$$c(n, p, \Omega) = C(n) \text{diam}(\Omega)^p.$$

This completes the proof. □

7 Problem 7

Theorem 7.1. *Let $\Omega \subset \mathbb{R}^n$ be open, bounded and C^1 . For every $1 \leq p < \infty$, there exists $c = c(n, p, \Omega) > 0$ such that*

$$\int_\Omega |u - (u)_\Omega|^p \leq c \int_\Omega |\nabla u|^p$$

for every $u \in W^{1,p}(\Omega)$.

Proof. We prove by contradiction. Assume no constant $c > 0$ exists. Then for every integer $k \geq 1$, choose $u_k \in W^{1,p}(\Omega)$ with:

$$\int_{\Omega} |u_k - (u_k)_{\Omega}|^p > k \int_{\Omega} |\nabla u_k|^p. \quad (17)$$

Define $w_k = u_k - (u_k)_{\Omega}$. Then:

$$\int_{\Omega} w_k = 0,$$

so (17) becomes:

$$\int_{\Omega} |w_k|^p > k \int_{\Omega} |\nabla w_k|^p. \quad (18)$$

Normalize:

$$v_k = \frac{w_k}{\|w_k\|_{L^p(\Omega)}}.$$

Then:

$$\|v_k\|_{L^p(\Omega)} = 1, \quad \int_{\Omega} v_k = 0.$$

Dividing (18) by $\|w_k\|_{L^p(\Omega)}^p$:

$$1 > k \int_{\Omega} |\nabla v_k|^p.$$

Thus:

$$\int_{\Omega} |\nabla v_k|^p < \frac{1}{k}.$$

So $\|\nabla v_k\|_{L^p(\Omega)} \rightarrow 0$ as $k \rightarrow \infty$. Also:

$$\|v_k\|_{W^{1,p}(\Omega)}^p = \|v_k\|_{L^p(\Omega)}^p + \|\nabla v_k\|_{L^p(\Omega)}^p \leq 1 + \frac{1}{k} \leq 2.$$

Hence $\{v_k\}$ is bounded in $W^{1,p}(\Omega)$.

Since Ω is bounded and C^1 , the Rellich-Kondrachov compactness theorem implies that $W^{1,p}(\Omega)$ is compactly embedded in $L^p(\Omega)$. Therefore, there exists a subsequence and a function $v \in L^p(\Omega)$ such that $v_k \rightarrow v$ strongly in $L^p(\Omega)$. In particular:

$$\|v\|_{L^p(\Omega)} = \lim_{k \rightarrow \infty} \|v_k\|_{L^p(\Omega)} = 1,$$

and because $\int_{\Omega} v_k = 0$ for all k , we also have $\int_{\Omega} v = 0$.

Moreover, for any test function $\phi \in C_c^\infty(\Omega)$ and any $i = 1, \dots, n$,

$$\int_{\Omega} v \partial_i \phi \, dx = \lim_{k \rightarrow \infty} \int_{\Omega} v_k \partial_i \phi \, dx = - \lim_{k \rightarrow \infty} \int_{\Omega} \partial_i v_k \phi \, dx.$$

Since $\|\partial_i v_k\|_{L^p(\Omega)} \leq \|\nabla v_k\|_{L^p(\Omega)} \rightarrow 0$, the right-hand side tends to zero. Hence:

$$\int_{\Omega} v \partial_i \phi \, dx = 0 \quad \text{for all } \phi \in C_c^\infty(\Omega).$$

This means v has weak derivative zero. Therefore v is constant on each connected component of Ω . Since Ω is connected, v is constant everywhere. Together with $\int_{\Omega} v = 0$, this forces $v = 0$ almost everywhere, contradicting $\|v\|_{L^p(\Omega)} = 1$.

Hence the assumption was false and there exists $c = c(n, p, \Omega) > 0$ such that:

$$\int_{\Omega} |u - (u)_{\Omega}|^p \leq c \int_{\Omega} |\nabla u|^p.$$

This completes the proof. $\square$

8 Problem 8

Proof. As $W^{1,\infty}(\Omega) \subset W^{1,p}(\Omega)$ for any $n < p < \infty$, according to the Sobolev embedding theorem:

Theorem 8.1. *Let $\Omega \subset \mathbb{R}^n$ be open, bounded and smooth and let $n < p < \infty$. Then $W^{1,p}(\Omega)$ continuously embeds into $C^{0,\tau}(\overline{\Omega})$ for every $0 \leq \tau \leq 1 - \frac{n}{p}$.*

Hence, for any $x, y \in \overline{\Omega}$:

$$|u(x) - u(y)| \leq c|x - y|^{1-\frac{n}{p}} \|u\|_{W^{1,p}(\Omega)}.$$

Letting $p \rightarrow \infty$ and noting that:

$$\lim_{p \rightarrow \infty} \|u\|_{W^{1,p}(\Omega)} = \|u\|_{W^{1,\infty}(\Omega)},$$

we obtain:

$$[u]_{C^{0,1}(\overline{\Omega})} := \sup_{x,y \in \overline{\Omega}} \frac{|u(x) - u(y)|}{|x - y|} \leq c \|u\|_{W^{1,\infty}(\Omega)}.$$

So we have established the continuous embedding:

$$W^{1,\infty}(\Omega) \subset C^{0,1}(\overline{\Omega}). \quad (19)$$

Now prove the reverse inclusion. By Rademacher's theorem, every Lipschitz function $u : \Omega \rightarrow \mathbb{R}$ is differentiable almost everywhere, and

$$|\nabla u(x)| \leq [u]_{C^{0,1}} \quad \text{for almost every } x.$$

Hence $\partial_i u \in L^\infty(\Omega)$ and $u \in L^\infty(\Omega)$. Since u is a.e. differentiable and bounded, integration by parts is justified:

$$\int_{\Omega} u \partial_i \phi = - \int_{\Omega} \partial_i u \phi \quad \forall \phi \in C_c^\infty(\Omega).$$

Thus the classical derivative equals the weak derivative. So:

$$u \in L^\infty(\Omega), \quad \nabla u \in L^\infty(\Omega).$$

Therefore $u \in W^{1,\infty}(\Omega)$ and:

$$\|u\|_{W^{1,\infty}} \leq \|u\|_{L^\infty} + [u]_{C^{0,1}}.$$

Hence:

$$C^{0,1}(\overline{\Omega}) \subset W^{1,\infty}(\Omega), \quad (20)$$

with equivalent norms:

$$\|u\|_{W^{1,\infty}} \cong \|u\|_{L^\infty} + [u]_{C^{0,1}}.$$

Combining (19) and (20):

$$\boxed{W^{1,\infty}(\Omega) = C^{0,1}(\overline{\Omega}).}$$

$\square$

9 Problem 9

Proof. Let $A \subset \mathbb{R}^n$ be measurable with $0 < |A| < \infty$. Let $f \in L^p(A)$, $(f)_A = \frac{1}{|A|} \int_A f$, and $\lambda \in \mathbb{R}$ arbitrary.

First decompose:

$$f - (f)_A = (f - \lambda) + (\lambda - (f)_A). \quad (21)$$

Using Young's inequality $|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p)$ for $p \geq 1$:

$$|f - (f)_A|^p \leq 2^{p-1}(|f - \lambda|^p + |\lambda - (f)_A|^p). \quad (22)$$

Integrating (22) over A :

$$\int_A |f - (f)_A|^p \leq 2^{p-1} \left(\int_A |f - \lambda|^p + |A| |\lambda - (f)_A|^p \right). \quad (23)$$

Applying Jensen's inequality:

$$|\lambda - (f)_A| = \left| \frac{1}{|A|} \int_A (\lambda - f(x)) dx \right| \leq \frac{1}{|A|} \int_A |f - \lambda| dx. \quad (24)$$

Applying Hölder's inequality to (24):

$$\frac{1}{|A|} \int_A |f - \lambda| \leq \left(\frac{1}{|A|} \int_A |f - \lambda|^p \right)^{1/p}. \quad (25)$$

From (24) and (25):

$$|\lambda - (f)_A|^p \leq \frac{1}{|A|} \int_A |f - \lambda|^p. \quad (26)$$

Multiplying (26) by $|A|$:

$$|A| |\lambda - (f)_A|^p \leq \int_A |f - \lambda|^p. \quad (27)$$

Substituting (27) into (23):

$$\int_A |f - (f)_A|^p \leq 2^{p-1} \left(\int_A |f - \lambda|^p + \int_A |f - \lambda|^p \right) = 2^p \int_A |f - \lambda|^p.$$

Thus:

$$\boxed{\int_A |f - (f)_A|^p \leq 2^p \int_A |f - \lambda|^p.}$$

Dividing by $|A|$ gives the averaged form. This proves the desired inequality. $\square$